Week5 TA Problems Projection

15 problems with worked solutions

Approx. 80 minutes: allow discussion and board work.

Each problem slide has intentional blank space; advance for the solution.

Problem 1: Two-line system

Solve $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$.

Work here before advancing.

Solution 1: Two-line system

$R_2 \leftarrow R_2 - 3R_1$ gives $-5y = -13$, so $y = 13/5$, $x = 9/5$.

Problem 2: Parallel or coincident?

Classify $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ c \end{bmatrix}$ for $c = 3$ and $c = 4$.

Work here before advancing.

Solution 2: Parallel or coincident?

Row 1 equals twice row 2 on the left. For $c = 3$ infinitely many solutions; for $c = 4$ no solution.

Problem 3: One elimination step

Eliminate below the first pivot in $\begin{bmatrix} 3 & 2 \\ 9 & 8 \end{bmatrix}$.

Work here before advancing.

Solution 3: One elimination step

Multiplier 3. $R_2 \leftarrow R_2 - 3R_1$ gives $\begin{bmatrix} 3 & 2 \\ 0 & 2 \end{bmatrix}$.

Problem 4: Triangular solve

$$\text{Solve } \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 6 \end{bmatrix}.$$

Work here before advancing.

Solution 4: Triangular solve

$z = 2$, $2y - 4 = 0 \Rightarrow y = 2$, $x + 4 + 2 = 8 \Rightarrow x = 2$.

Problem 5: A full three-variable system

Solve $\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 2 & 3 & 1 & 9 \\ 1 & 0 & 1 & 4 \end{array}\right]$.

Work here before advancing.

Solution 5: A full three-variable system

After $R_2 - 2R_1$ and $R_3 - R_1$: $\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & -1 & 1 & 1 \end{array}\right]$. Add row 2 to row 3: $2z = 4$.

Then $z = 2$, $y = 1$, $x = 2$.

Problem 6: Build E_{32}

Write the 3×3 elementary matrix for $R_3 \leftarrow R_3 + 5R_2$.

Work here before advancing.

Solution 6: Build E_{32}

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}.$$

Problem 7: Reverse the operation

Find the inverse of $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$.

Work here before advancing.

Solution 7: Reverse the operation

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}.$$

Problem 8: Transform a right-hand side

Apply $E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ to $\mathbf{b} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$.

Work here before advancing.

Solution 8: Transform a right-hand side

$E\mathbf{b} = \begin{bmatrix} 3 \\ -1 \\ 7 \end{bmatrix}$. The second component becomes $5 - 2(3) = -1$.

Problem 9: Row swap matrix

Find P swapping rows 2 and 3 of a 3×3 matrix.

Work here before advancing.

Solution 9: Row swap matrix

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } P^2 = I.$$

Problem 10: A system requiring a swap

Solve $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 14 \end{bmatrix}$.

Work here before advancing.

Solution 10: A system requiring a swap

Swap rows: $2x + 3y = 14$, $y = 4$. Hence $x = 1$.

Problem 11: Free variable

Solve $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$.

Work here before advancing.

Solution 11: Free variable

Set $z = t$, then $y = 3 - t$ and $x = 1$. Solutions $\begin{bmatrix} 1 \\ 3 - t \\ t \end{bmatrix}$.

Problem 12: Inconsistent row

Explain why $\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{array}\right]$ has no solution.

Work here before advancing.

Solution 12: Inconsistent row

Last row reads $0 = 1$, an impossibility.

Problem 13: Compare EP and PE

For $E = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, compute both products.

Work here before advancing.

Solution 13: Compare EP and PE

$EP = \begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}$, $PE = \begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix}$. They differ.

Problem 14: Column space test

Can $\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ be solved?

Work here before advancing.

Solution 14: Column space test

No. All column combinations have second component twice the first, but $5 \neq 2(2)$.

Problem 15: Challenge: build a permutation

Construct P that sends rows (R_1, R_2, R_3) to (R_3, R_1, R_2).

Work here before advancing.

Solution 15: Challenge: build a permutation

$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. It permutes rows by left multiplication.